DAEN/ISEN 427 In class assignment

Due: Thu Sep 11, 2025 (in class)

Problem

In an attempt to catch the terrorists, a federal agency installs an alarm system with a surveillance camera and automatic facial recognition software to identify 10 known terrorists in a city of 1 million inhabitants (let us assume that all people present in the city are inhabitants).

The system:

- Fails to detect a terrorist 0.5% of the time (false negative rate).
- Incorrectly flags a non-terrorist 0.5% of the time (false positive rate).

A survey show that when someone triggers the alarm, people often believe with more than 99% certainty that the individual is a terrorist.

Your assignment

1. Compute the probability that a randomly selected individual is a terrorist.

Answer: There are 10 terrorists out of 1,000,000 inhabitants:

$$P(\text{terrorist}) = \frac{10}{1{,}000{,}000} = 0.00001 \quad (0.001\%)$$

The chance that a randomly chosen individual is a terrorist is extremely small (1 in 100,000)

2. Calculate the probability that someone who triggered the alarm is a terrorist.

Answer: We use Bayes' theorem. So let:

$$P(T) = 0.00001, \quad P(NT) = 1 - 0.00001 = 0.99999$$

$$P(\text{alarm}|T) = 0.995, \quad P(\text{alarm}|NT) = 0.005$$

Then:

$$P(T|\text{alarm}) = \frac{P(\text{alarm}|T)\,P(T)}{P(\text{alarm}|T)\,P(T) + P(\text{alarm}|NT)\,P(NT)}$$

Substituting:

$$P(T|\text{alarm}) = \frac{0.995 \cdot 0.00001}{0.995 \cdot 0.00001 + 0.005 \cdot 0.99999} \approx 0.00198607 \quad (0.199\%)$$

So even if the alarm triggers, the probability the person is truly a terrorist is about 0.2%, still very small (1 in 500)

3. If the city wanted at least a 50% chance that a flagged person is a terrorist, what maximum false positive rate would the system need?

Answer: We require:

$$0.5 \leq \frac{0.995 \cdot 0.00001}{0.995 \cdot 0.00001 + \text{FPR} \cdot 0.99999}$$

Solving for FPR:

$$\text{FPR} \leq \frac{0.995 \cdot 0.00001}{0.99999} \approx 0.00000995 \quad (0.00099\%)$$

Thus the false positive rate would need to be less than 1 in 100,503

4. Explain why improving sensitivity (reducing false negatives) often worsens specificity (increasing false positives).

Answer: Sensitivity measures the ability to detect terrorists. Specificity measures the ability to avoid flagging innocents. If the detection threshold is lowered to catch more terrorists (higher sensitivity), more non-terrorists are also flagged (lower specificity). Both measures trade off against one another because they depend on the same threshold.

5. In the context of terrorism detection, which is more critical to minimize: false positives or false negatives?

Answer:

- **False negatives** (missed terrorists) risk real harm if an attack occurs.
- **False positives** (innocents flagged) cause wasted resources, stigmatization, and undermine of civil liberties.

(Possible answer) Given the extremely low prevalence of terrorists, minimizing false positives is more critical to prevent overwhelming the system and unjustly accusing innocents. At the same time, a balance must be maintained to avoid missing actual threats.

6. Briefly discuss: Would the use of such surveillance systems likely meet legal standards? How should societies weigh this tradeoff?

For reference: Apple Face ID is designed to be inaccurate in less than 1 in 1,000,000 and Apple Touch ID to be inaccurate in less than 1 in 50,000. The theoretical probability of someone randomly guessing a four-digit passcode on the first try is 1 in 10,000. For a six-digit passcode, it is 1 in 1,000,000. In theory, this makes a six-digit passcode roughly as secure as Face ID, but there are other considerations (iPhones limit the number of failed passcode attempts, so manual brute-force attacks don't work on these devices. However, automated brute-force attacks are possible if the attacker has specialized tools).

Answer: *(Possible answer)* Such system are unlikely to be justifiable as the harms to civil liberties outweigh the minimal security benefits. The reliability of the system (predictive value) seems to not satisfy standards of *reasonable suspicion* or *probable cause*.

Suppose now that an inhabitant triggers the alarm. Someone making the base rate fallacy might conclude that there is a 99.5% chance the detected person is a terrorist. At first glance this seems reasonable (the system is very accurate) but it is actually bad reasoning. A correct calculation shows the probability is not near 99% but closer to 0.2% (about 1 in 500). The mistake comes from confusing two different probabilities:

- $P(\text{alarm}|T)$: the probability the alarm fails to trigger given the person is a terrorist (false negative rate = 0.5%).
- $P(NT|\text{alarm})$: the probability the person is not a terrorist given that the alarm triggers.

These are completely different quantities. One cannot be inferred from the other. To see why, imagine the system installed in a different city with no terrorists at all. The alarm would still sound for 0.5% of the population (false positives), but never for a terrorist. In that case:

- $P(T|\text{alarm}) = 0\%$.
- Yet the false negative rate $P(\text{not alarm}|T)$ is not even defined since there are no terrorists.

This shows how unrelated the two failure rates are.